

Program : Diploma in Computer Application and Business Management	
Course Code : 1254	Course Title: Foundations in Science & Technology
Semester : 1	Credits: No Credit
Course Category: Foundation Course	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To provide the students an understanding with the methods of science and scientific knowledge, water treatment methods, measurement concepts and familiarization with the concept of atomic nucleus.

Course Prerequisites:

Topic	Course/Program name
Physics and Chemistry	Secondary School

Course Outcomes:

On completion of the course, students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Describe the methods of science and use scientific reasoning to solve problems	14	Applying
CO2	Apply the fundamentals of analytical chemistry and understand appropriate water treatment methods	16	Applying
CO3	Utilize the measurement concepts and the laws of motion	15	Applying
CO4	Explain the concept and force of atomic nucleus	13	Understanding
	Series Exam	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2					3		
CO3	3						
CO4					3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe the methods of science and use scientific reasoning to solve problems		
M1.01	Know the concept of methods of science	5	Remembering
M1.02	Understand the nature of scientific knowledge	4	Understanding
M1.03	Use scientific reasoning to solve problems	3	Applying
M1.04	Explain modern developments in science and technology	2	Understanding
Contents: The method of science concept, Steps in scientific method i.e. observation, hypothesis, experiments and apparatus, development of laws, models and theories, examples to illustrate the method of science - preparation of food by plants, invention of barometer, burning of candle and conservation of mass. The nature of scientific knowledge, dynamic, never complete, repeatable, verifiable, open to change and self-correcting nature of scientific knowledge. Scientific nature to problem solving, use of scientific reasoning and logic to solve problems, the strategy of science, use of scientific achievements for better living and removal of ignorance. Modern Developments in Science and Technology - Introduction to Laser, Fibre Optics, Space Technology			
CO2	Apply the fundamentals of analytical chemistry and understand appropriate water treatment methods		
M2.01	Make use of the knowledge of concentration of solutions and volumetric analysis as a quantitative analysis in the field of science, engineering and technology	6	Applying

M2.02	Apply the physical concepts related to pH and develop the skill of solving problems	4	Applying
M2.03	Explain different types of hardness of water and the methods of removal of hardness	3	Understanding
M2.04	Explain various steps involved in municipal water treatment	3	Understanding
	Series Test – I	1	
Contents: Solution - idea of solute, solvent and solution, methods to express the concentration of solution - molarity, normality and ppm. Simple problems based on molarity and normality. Ionic Product of water - pH and pOH scale Definition - relation between pH and pOH - Simple problems based on pH. Applications of pH - Buffer solution - definition - classification of Buffers - Acidic and Basic Buffers. Volumetric analysis - Titration - end point - indicators - pH range of indicators - choice of indicators in titration - Principle of Volumetric analysis (Normality equation only). Simple problems based on normality equation Classification of soft and hard water, salts causing water hardness. Cause of poor lathering of soap in hard water. Problems caused by the use of hard water in boiler (scale and corrosion), Water softening techniques - soda lime process and ion exchange method. Potable water and its characteristics, Municipal water treatment (in brief only) - sedimentation, coagulation, filtration, sterilization - chlorination, bleaching, UV radiation. Flow chart for municipal supply of potable water.			
CO3	Utilize the measurement concepts and the laws of motion		
M3.01	Explain about the different concepts used in measuring	6	Understanding
M3.02	Use the equations of motions to solve problems	5	Applying
M3.03	Understand Newton's Laws and their applications	4	Applying
Contents: Measuring Units - Systems of Units, Fundamental and derived Units - s. i. units, Dimensions - Dimensions for different physical quantities, dynamics of particles and newton's laws of motion, displacement, velocity, acceleration, uniform velocity, and uniform acceleration, Equations of Motion, Solve simple problems on equation of motion, Motion under gravity and 'g'. Newton's first law-inertia, force concept and momentum, Newton's second law - derive expression $f=mg$, Newton's third law - quote examples in daily life, conservation of momentum, recoil of gun			
CO4	Explain the concept and force of atomic nucleus.		
M4.01	Understand the structure of nucleus and nuclear force	3	Understanding

M4.02	Describe radioactivity	3	Understanding
M4.02	Explain nuclear fission and fusion	4	Understanding
M4.03	Understand nuclear holocaust and the hazards caused by radiation	3	Understanding
	Series Test – II	1	
Contents: Structure of Nucleus, Nuclear Force, Radioactivity, Nuclear Energy, difference between Nuclear Fission and Nuclear Fusion, Nuclear Holocaust, Radiation Hazards			

Text / Reference:

T/R	Book Title/Author
R1	FST-I-Foundation course in Science and Technology-Block 1 Unit I, Block 2 Unit 1, Block 7 Unit 1 , IGNOU, Maidan Garhy, New Delhi
R2	N. R. Rao, Understanding Chemistry , Universities Press (India) Pvt. Ltd., 2011.
R3	R.V.G Menon, History of Science and Technology , Pearson
R4	Arthur Beiser, Modern Physics , Tata McGraw
R5	William E. Stephens, Nuclear Fission and Atomic Energy , Inman Press 2007

Online Resources:

Sl.No	Website Link
1	https://ignoubcom.wordpress.com/study-material/fst1sm/
2	https://www.wastewaterelearning.com/elearning/ (Water Treatment)
3	https://physics.info/motion/
4	https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Supplemental_Modules_(Physical_and_Theoretical_Chemistry)/Nuclear_Chemistry/Fission_and_Fusion/Fission_and_Fusion